



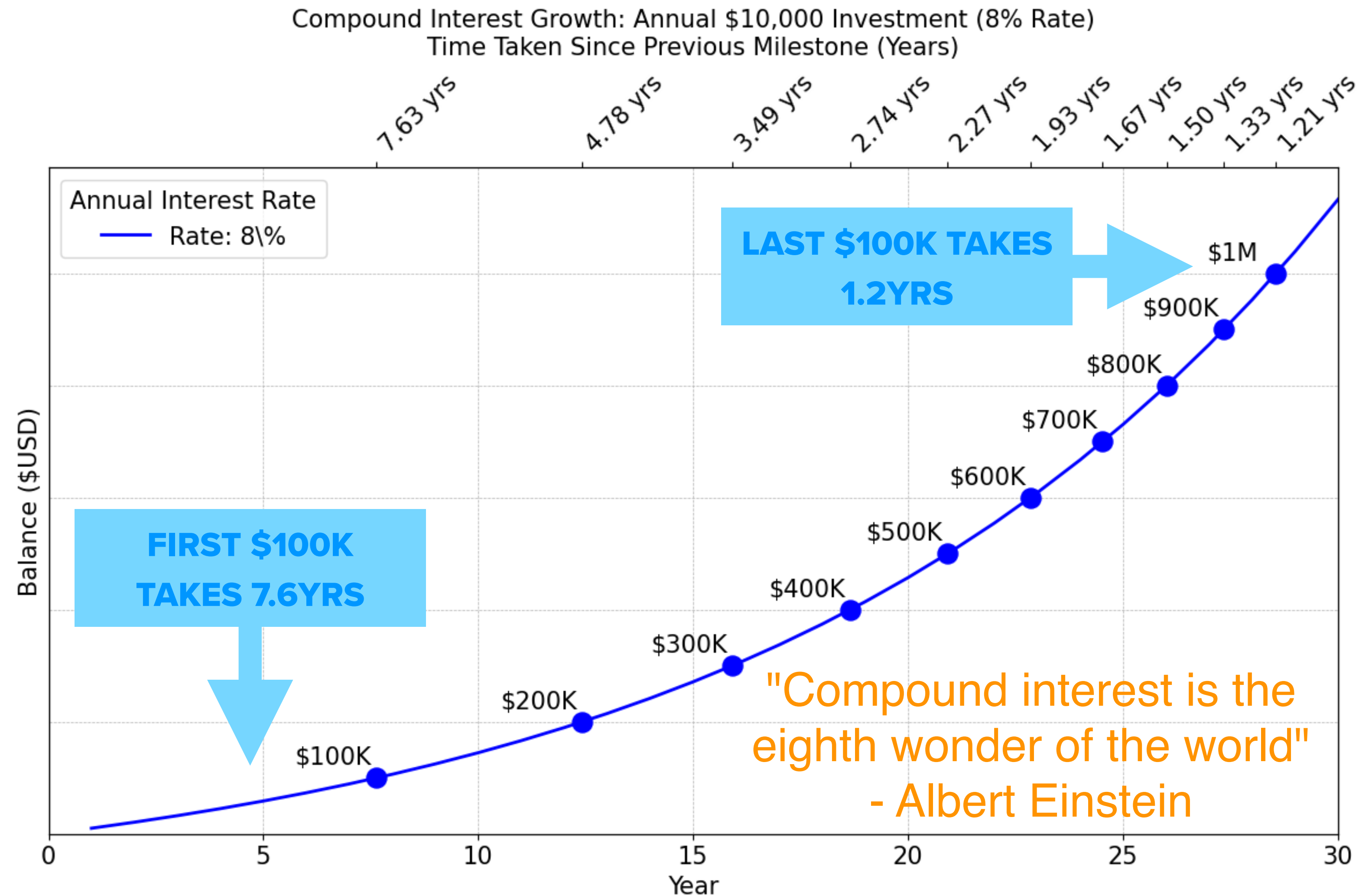
LUIS ROSARIO 2024

DISCLAIMER

- I am **NOT** endorsed by anyone and I am not promoting anyone.
- This is for educational and entertainment purposes only and **NOT** financial advice.
- I am an Engineer, not a CPA or CFA.

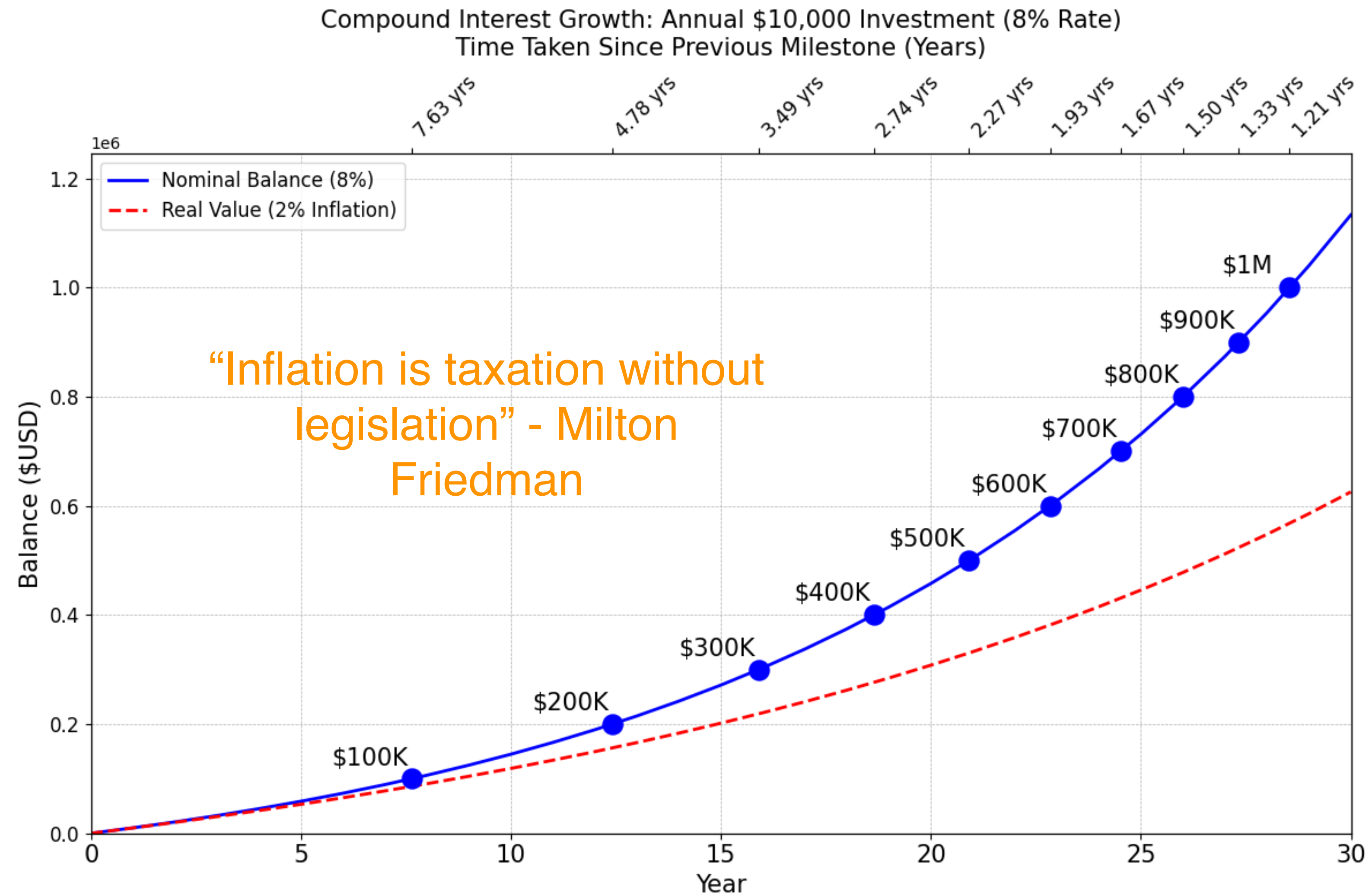
POWER OF COMPOUNDING INTEREST

- Most people earning **\$85,000** a year believe that their only shot at becoming a millionaire is to win the lottery. However, the power of compound interest illustrates how anyone can slowly transform investment in a large fortunes over time.
- How much money do you need to invest to reach **\$1M** if you start investing when you are 10 years old and want to use the money when you are 65 years old? Answer



POWER OF INFLATION

- Money loses its purchasing power over time (aka is worth less)
- Saving money in the bank/under the mattress is not a good idea due to inflation
- Using the CPI, what cost \$1 in 1970 took \$6.19 to buy in 2014. A six-fold increase. In the same time period, a 4-year state school college education went from \$4,800 to \$160,000. A 33-fold increase.



INVESTMENT VEHICLES

■ Work Retirement

- 401k
- TSP
- 403b
- 457

■ Health Savings

- Flexible Saving Account
- Limited Flexible Saving Account
- Health Saving Account

■ Personal Retirement

- IRA
 - Traditional
 - ROTH
- Brokerage Accounts
- Crypto
- Options
- Annuities
- Life Insurance
- Stocks

■ 529 Saving Plans

- Money Market Account
- Money Market Fund
- Certificates of Deposit
- Bonds
- TIPS
- I Bonds
- Cash Mgmt. Account

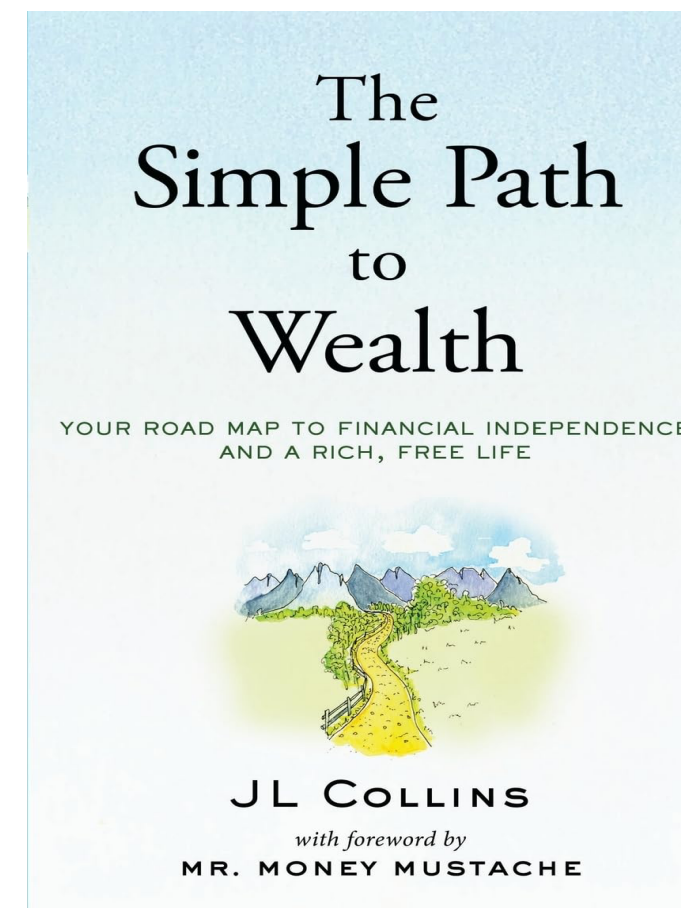
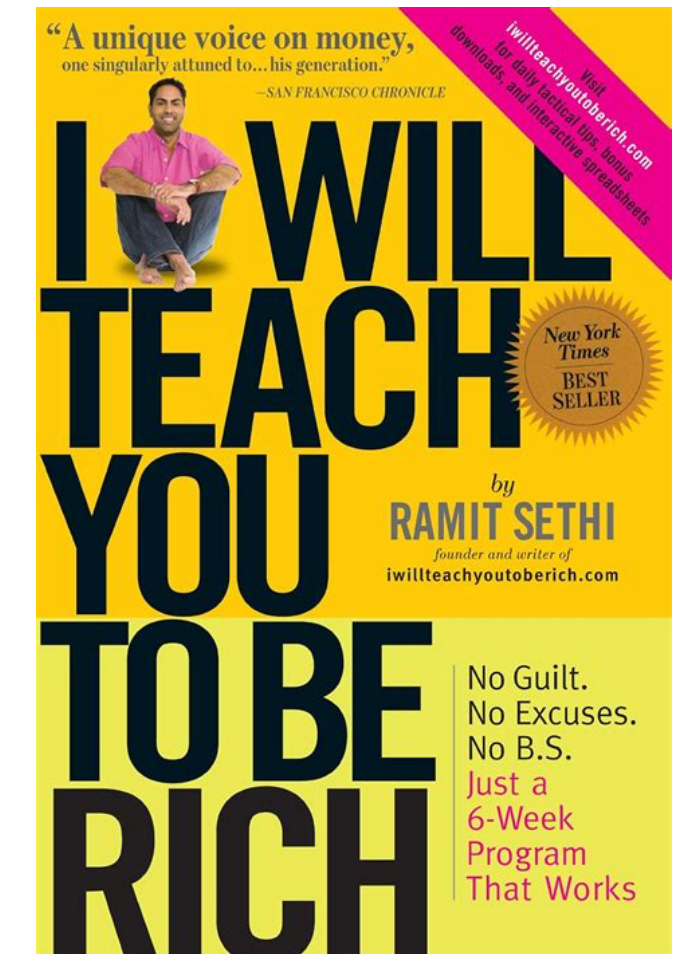
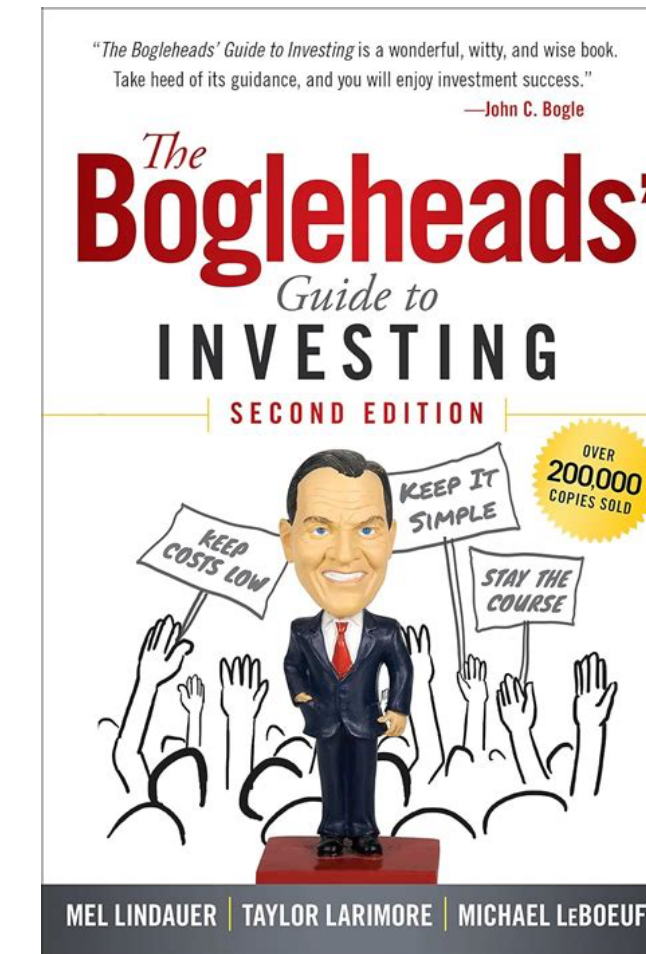
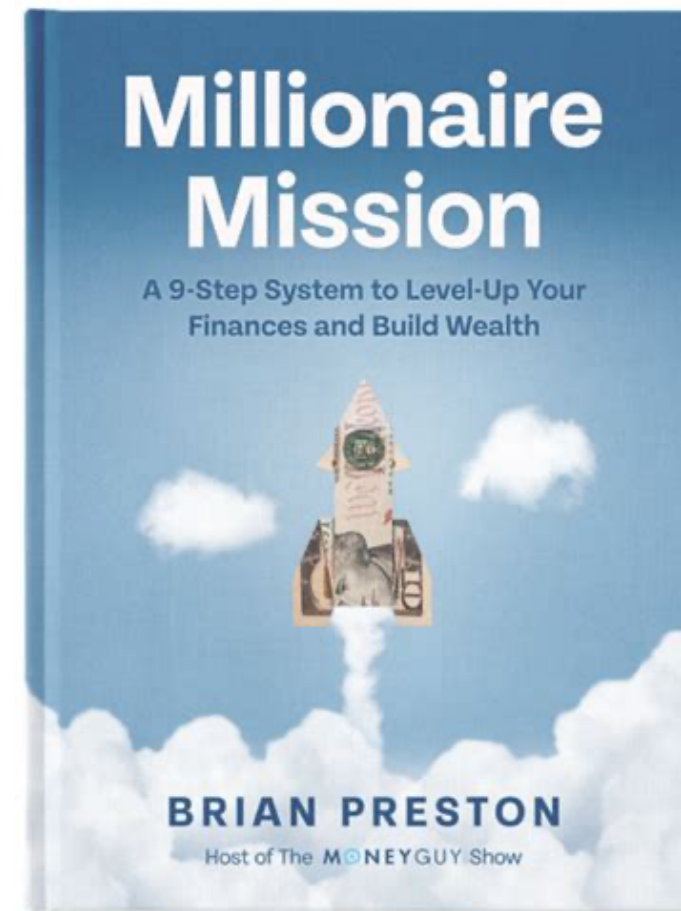
INVESTMENT ACCOUNTS SUMMARY

	Maximum (2026)***	Employer Contribution	RMD	Income Limit	Access Age	Taxes
Traditional 401k	\$25000	Yes	Yes	No	59 1/2	Deferred
Roth 401k	\$25000	No	No	No	59 1/2	2x Tax free
Traditional IRA	\$7500	No	Yes	Yes	59 1/2	Deferred
Roth IRA	\$7500	No	No	Yes	59 1/2	2x Tax Free
HSA	\$4500*/\$8550**	Yes	No	No	Retroactive	3x Tax Free
LP-FSA	\$3400	No	No	No	Instant	2x Free
529	\$19k*/\$38k**	No	No	Yes	Instant	2x Free
<u>Brokerage Acc.</u>	Unlimited	No	No	No	Instant	Taxable
Cash Mgmt.	Unlimited	No	No	No	Instant	Taxable

*For a single under 55, **Married under 55, ***Age and civil status will vary contributions

HOW MUCH TO INVEST? SAVING...

- Streamline Expenses
 - Improve Credit Score
 - Refinance Car, Mortgage with lower rates
 - Debt Consolidation
 - Pay Off Debt
- Don't carry a credit card balance
- Shop for insurance
- Goals, Health and Age
 - Make up period
 - Accumulation/Preservation

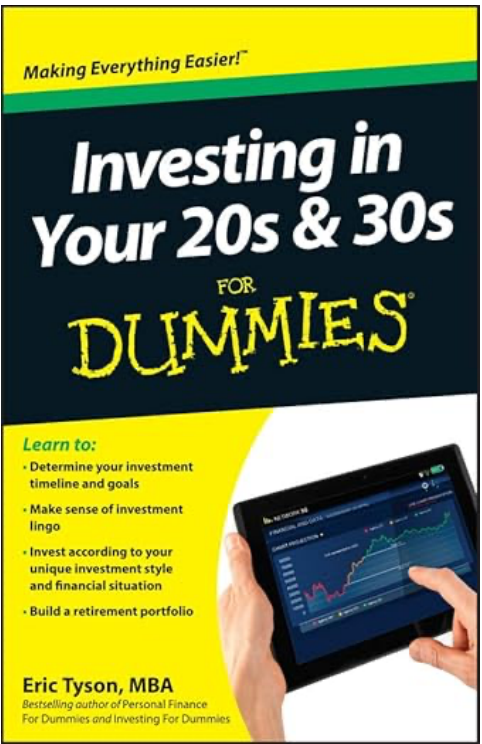
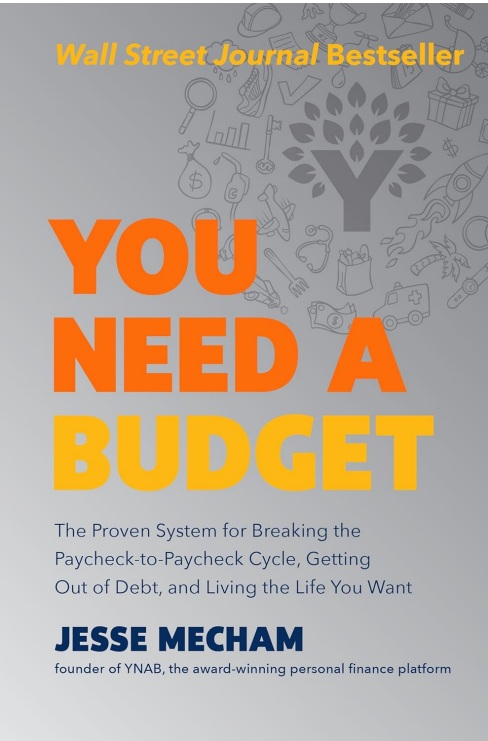
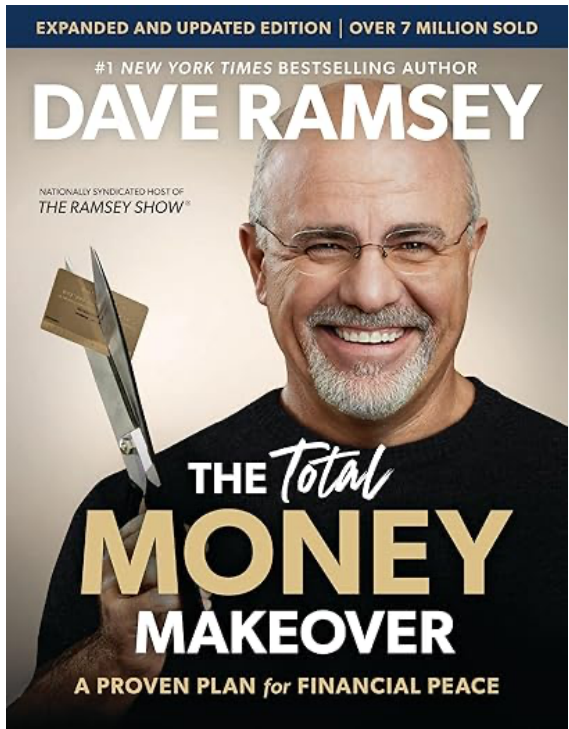


“If you intend to achieve financial freedom, you are going to have to think differently. It starts by recognizing that debt should not be considered normal. It should be recognized as the vicious, pernicious destroyer of wealth-building potential it truly is. It has no place in your financial life... the more debt you carry the more of your income is devoured by interest payments.” - JL Collin

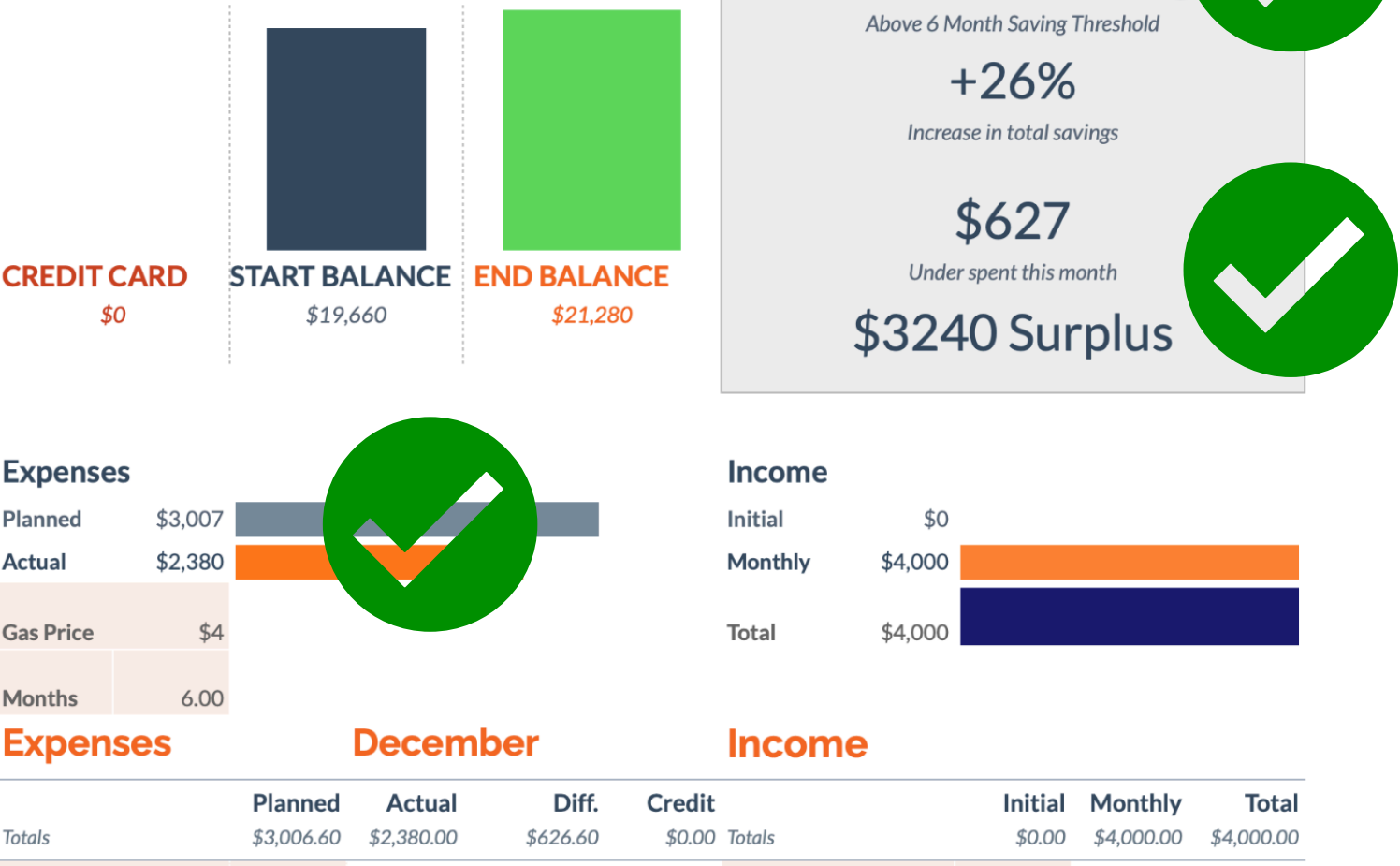
IT IS NOT HOW MUCH YOU MAKE BUT HOW MUCH YOU CAN SAVE.

HOW MUCH TO INVEST? BUDGETING...

- Budget Apps
- Track Expenses (i.e. subscriptions, cell-phones)
- Zero-Based Budgeting and cash flow
- Avoid Lifestyle Creep
 - Downsize house or car and memberships
 - Use a library card for audio books and movies
- Tax efficiency and tax planning
- Avoid fees (late, bank, credit card, utilities, annual fees) using Auto-payment
- All in one app (Saving Planner, Cash Flow, Budgeting, etc), WPCU, EMPOWER, Chase, etc...



Monthly Budget



“SUPPOSE ONE OF YOU WANTS TO BUILD A TOWER (STABLE FINANCIAL FUTURE). WON'T YOU FIRST SIT DOWN AND ESTIMATE THE COST TO SEE IF YOU HAVE ENOUGH MONEY TO COMPLETE IT?”

HOW TO INVEST? ORDER OF INVESTMENT

- Both frameworks have their merits and place in investing. Depending of your mindset and financial situation you might need the Baby Steps and could transition to the FOO later.
- Watch some videos on comparing these two methods to get more familiar with the thought process of how to plan for the future
 - Video 1
 - Video 2
 - Video 3



Caution one size does not fit all. Low interest rate debt payment vs Investment

A GOOD MAN LEAVES AN INHERITANCE TO HIS CHILDREN'S CHILDREN... -PROVEEBS 13:22

WHERE TO INVEST?

- Account Types: Taxable, Tax Deferred, Tax-Free Accounts
 - IRA, 401k, 529, Cash Mgmt. Acc., Brokerage Account, ...
- Brokerage Firms: Vanguard, Fidelity, Charles Schwab, eTrade, RobinHood, ...
 - Assets Under Management, Lowest Cost and Tax Efficiency
- Management Types: Active, Passive (Index) Products
 - Passive is cheaper and tends to out perform active management
 - Investment products: Mutual Funds, ETF, Stocks, Bonds, ...
 - ETFs: VOO, SPY, FXIAX, ...
 - Stocks: ARM, NVDA, AAPL, ...
 - Bonds: Junk bonds, TIPS, I Bonds, ...

CASH MANAGEMENT ACCOUNT

	CD	High Yield	Cash Mgmt
Liquidity	Limited	High	High
Minimum Req	\$500-\$1000	\$5	\$0
Inflation Risk	Not inflation protection	Promotional Rates	Moves with Inflation
Opportunity Cost	Locked	Not Locked	Not Locked
Flexibility	Less Flexible	High Flexibility	High Flexibility
Taxes	Ordinary Tax	Ordinary Tax	Qualified Dividends

DETAILED COMPARISON (RECOMMEND READ) CURRENTLY MUTUAL FUNDS IN A CASH MANAGEMENT ACCOUNT ARE THE BEST PLACE TO HAVE CASH BUT THAT IS NOT ALWAYS THE CASE.



<https://investor.vanguard.com/accounts-plans/vanguard-cash-plus-account>
<https://www.fidelity.com/spend-save/fidelity-cash-management-account/overview>

CASH MANAGEMENT ACCOUNT

Vanguard Cash Plus

- Next Day Transfers
- Pay bills
- FDIC and SPIC Insured
- Most Tax Efficient Mutual Funds
- Best Rates
- Paypal and Venmo
- Lowest Expense Ratios (0.07-0.12%)
- No Fees*



Comparison of Mutual Funds

Fidelity Cash Management

- Next Day Transfers
- Pay bills
- FDIC and SPIC Insured
- Debit Card
- Free Check-writing
- Good Rates
- Paypal and Venmo
- Highest Expense Ratios (0.42-0.47%)
- No Fees*



Comparison of Three Accounts

Schwab Bank Investor Checking

- Next Day Transfers
- Pay bills
- FDIC Insured
- Debit Card
- Free Check-writing
- Lower Rates
- Zelle, Apple Pay, Google Pay, Paypal and Venmo
- Modest Expense Ratios (0.34%)
- No Fees*
- No Overdraft Fees



AMERICANS HAVE AN ESTIMATED \$1.7 TRILLION SITTING IN SAVINGS ACCOUNTS THAT COULD BE EARNING WAY MORE INTEREST.

401K, TSP AND 403B

— Benefits

- Contribution Matching
- High Contribution Limits
- Tax Deferred (Tax Reduction)
Traditional
- Roth Conversion

— Cons

- Limited Investment Selection
- Penalties if retrieved before 59 1/2
- Traditional has Required Minimum Distribution at 72
- Some Have a Vetting Period

BROKERAGE ACC.	INCOME	TAX	INVEST	TAX	INVEST	TAX	INVEST	RETRIEVE	TAX
TRADITIONAL 401K	INCOME	INVEST	INVEST	INVEST	INVEST	INVEST	INVEST	RETRIEVE	TAX
ROTH 401K	INCOME	TAX	INVEST	INVEST	INVEST	INVEST	INVEST	RETRIEVE	NO TAX

<https://www.irs.gov/newsroom/401k-limit-increases-to-24500-for-2026-ira-limit-increases-to-7500>

<https://www.youtube.com/watch?v=sFaYjRyoFXw>

HOW MUCH TO INVEST?

- Evaluate Employer Matching
- Careful to not over contribute

Monthly Contributions								
	10.000000	12.000000	13.000000	14.000000	15.000000	16.000000	17.000000	18.000000
Income								
156,000.00	1,300.00	1,560.00	1,690.00	1,820.00	1,950.00	2,041.67	2,041.67	2,041.67
208,000.00	1,733.33	2,041.67	2,041.67	2,041.67	2,041.67	2,041.67	2,041.67	2,041.67
312,000.00	2,041.67	2,041.67	2,041.67	2,041.67	2,041.67	2,041.67	2,041.67	2,041.67
416,000.00	2,041.67	2,041.67	2,041.67	2,041.67	2,041.67	2,041.67	2,041.67	2,041.67
520,000.00	2,041.67	2,041.67	2,041.67	2,041.67	2,041.67	2,041.67	2,041.67	2,041.67
624,000.00	2,041.67	2,041.67	2,041.67	2,041.67	2,041.67	2,041.67	2,041.67	2,041.67
Annual Contributions								
	10.000000	12.000000	13.000000	14.000000	15.000000	16.000000	17.000000	18.000000
Income								
156,000.00	15,600.00	18,720.00	20,280.00	21,840.00	23,400.00	24,500.00	24,500.00	24,500.00
208,000.00	20,800.00	24,500.00	24,500.00	24,500.00	24,500.00	24,500.00	24,500.00	24,500.00
312,000.00	24,500.00	24,500.00	24,500.00	24,500.00	24,500.00	24,500.00	24,500.00	24,500.00
416,000.00	24,500.00	24,500.00	24,500.00	24,500.00	24,500.00	24,500.00	24,500.00	24,500.00
520,000.00	24,500.00	24,500.00	24,500.00	24,500.00	24,500.00	24,500.00	24,500.00	24,500.00
624,000.00	24,500.00	24,500.00	24,500.00	24,500.00	24,500.00	24,500.00	24,500.00	24,500.00

Annual Employer Contributions								
	1.000000	2.000000	3.000000	4.000000	5.000000	6.000000	7.000000	10.000000
Income								
156,000.00	1,560.00	3,120.00	4,680.00	6,240.00	7,800.00	9,360.00	10,920.00	15,600.00
208,000.00	2,080.00	4,160.00	6,240.00	8,320.00	10,400.00	12,480.00	14,560.00	20,800.00
312,000.00	3,120.00	6,240.00	9,360.00	12,480.00	15,600.00	18,720.00	21,840.00	31,200.00
416,000.00	4,160.00	8,320.00	12,480.00	16,640.00	20,800.00	24,960.00	29,120.00	41,600.00
520,000.00	5,200.00	10,400.00	15,600.00	20,800.00	26,000.00	31,200.00	36,400.00	47,500.00
624,000.00	6,240.00	12,480.00	18,720.00	24,960.00	31,200.00	37,440.00	43,680.00	47,500.00
Annual Total Contributions								
	11.000000	14.000000	16.000000	18.000000	20.000000	22.000000	24.000000	28.000000
Income								
156,000.00	17,160.00	21,840.00	24,960.00	28,080.00	31,200.00	33,860.00	35,420.00	40,100.00
208,000.00	22,880.00	28,660.00	30,740.00	32,820.00	34,900.00	36,980.00	39,060.00	45,300.00
312,000.00	27,620.00	30,740.00	33,860.00	36,980.00	40,100.00	43,220.00	46,340.00	55,700.00
416,000.00	28,660.00	32,820.00	36,980.00	41,140.00	45,300.00	49,460.00	53,620.00	66,100.00
520,000.00	29,700.00	34,900.00	40,100.00	45,300.00	50,500.00	55,700.00	60,900.00	72,000.00
624,000.00	30,740.00	36,980.00	43,220.00	49,460.00	55,700.00	61,940.00	68,180.00	72,000.00

HEALTH SAVING ACCOUNT

In order to contribute to an HSA, you have to be enrolled in an HSA-eligible health plan. Health savings accounts (HSAs) let you save and pay for qualified medical expenses with tax-free dollars (pre-tax, no tax on growth and no tax at

Every year, the Internal Revenue Service (IRS) sets the maximum that can be contributed to an HSA. For example, for the year 2026 the contribution limit is \$4,400 and your employer contributes \$1,000, you can contribute \$3,400—unless you're eligible for a catch-up contribution of \$1,000.

In 2026, you can contribute up to \$4,400 if you have health coverage just for yourself or \$8,750 if you have coverage for your family.

At age 55, individuals can contribute an additional \$1,000

HSA can be coupled with LP-FSA to get \$3,400 extra tax free money

HSA Tax Reduction		
2026 Single		Tax Saving
\$4,400.00	0.10%	\$440.00
\$4,400.00	0.12%	\$528.00
\$4,400.00	0.22%	\$968.00
\$4,400.00	0.24%	\$1056.00
\$4,400.00	0.32%	\$1408.00
\$4,400.00	0.35%	\$1540.00
\$4,400.00	0.37%	\$1628.00
2026 Married		Tax Saving
\$8,750.00	0.10%	\$875.00
\$8,750.00	0.12%	\$1050.00
\$8,750.00	0.22%	\$1925.00
\$8,750.00	0.24%	\$2100.00
\$8,750.00	0.32%	\$2800.00
\$8,750.00	0.35%	\$3062.50
\$8,750.00	0.37%	\$3237.50

HEALTH SAVING ACCOUNT

HSA	INCOME	INVEST	INVEST	INVEST	INVEST	INVEST	INVEST	RETRIEVE	NO TAX
TRADITIONAL	INCOME	INVEST	INVEST	INVEST	INVEST	INVEST	INVEST	RETRIEVE	TAX
ROTH	INCOME	TAX	INVEST	INVEST	INVEST	INVEST	INVEST	RETRIEVE	NO TAX

When you turn 65, you can use HSA money for ineligible expenses penalty free, though you'll have to pay income taxes. That is, you can do what ever you want with the money.

If you exceed the annual maximum contribution limit, you may face a 6% excess tax on your excess contributions in the year you over-contributed and in each year you fail to remove the excess contribution and its earnings. The excess contribution is also considered taxable income. If you correct the error before the tax filing deadline for the year, you may be able to avoid income tax and the excise tax for that year.

529 SAVINGS ACCOUNT

529 plans are tax-advantaged accounts designed specifically for education savings. 529 funds can be used for a wide range of education expenses, including college expenses at postsecondary schools nationwide, tuition for K-12 schools, certain apprenticeship costs, and student loan repayments.

The 529 account beneficiary can be changed to an eligible family member to use for their qualified education expenses.

Earnings grow tax deferred and withdrawals for qualified education expenses are are free from federal income tax and some state taxes.

You can gift money to an existing 529 plan account. That is, you can get a tax write-off for gifting money to someones 529 account.

Withdrawals from a 529 plan account can be taken at any time, for any reason. But, if the money is not used for qualified education expenses, federal income taxes may be due on any earnings withdrawn. A 10% federal penalty tax and possibly state or local tax can also be added. There are exceptions to the 10% penalty—for instance, if the beneficiary receives a scholarship or attends a US military academy. Any earnings would still be subject to federal income tax and any state and local taxes.

529 can be rolled-over to a ROTH IRA since 1/1/2024



ROTH INDIVIDUAL RETIREMENT ARRANGEMENT (IRA)

- Benefits
 - Unlimited Investment Selection
 - Tax Free Capital Gains
 - No Required Minimum Distribution
 - Investment can be retrieved before 59 1/2 without penalties
 - Back Door ROTH IRA
- Cons
 - Low Contribution Limits
 - Limited by Income Level

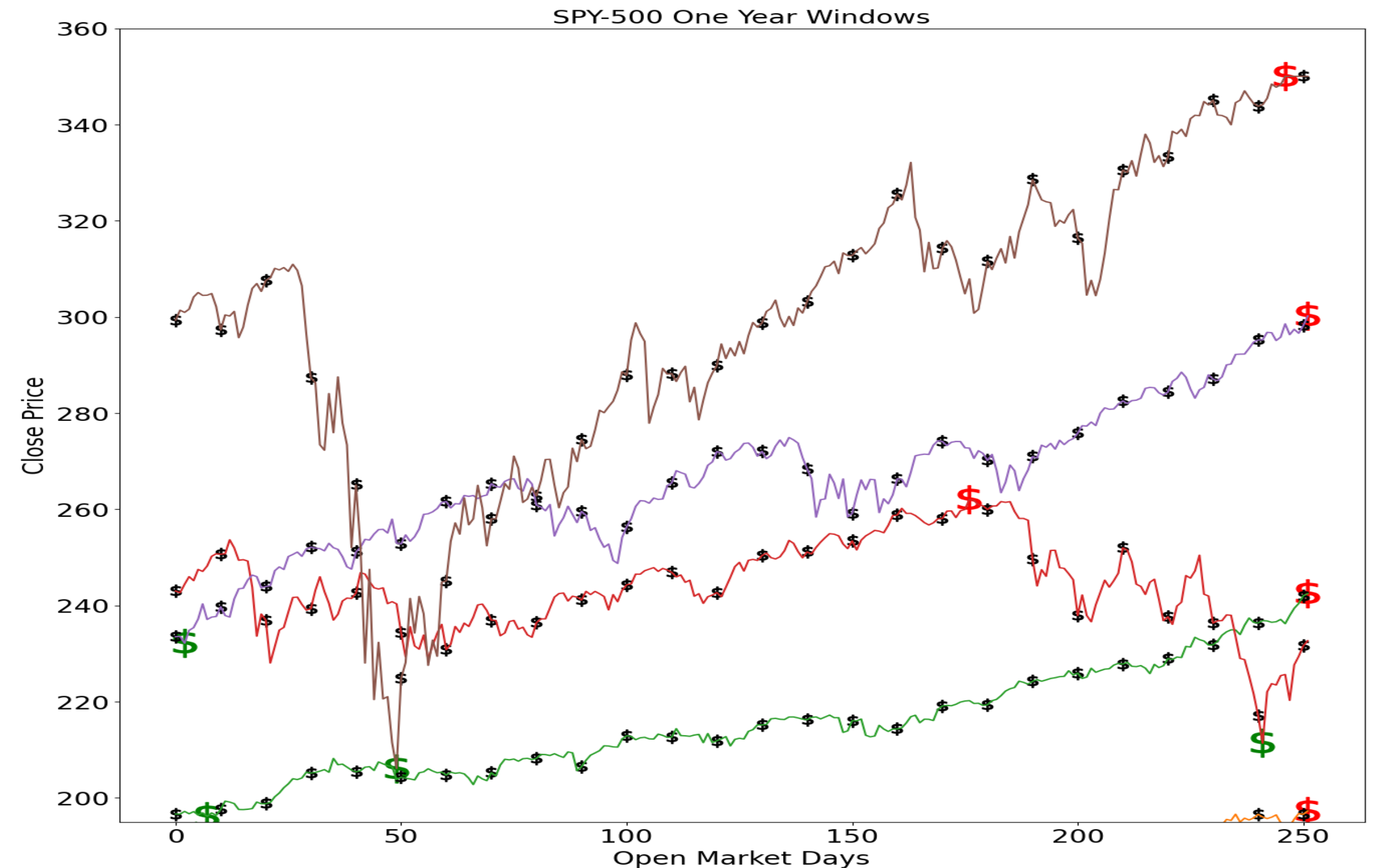
Roth IRA Contribution Limits (Tax Year 2025)

Single Filers (MAGI)	Married Filing Jointly (MAGI)	Married Filing Separately (MAGI)	Maximum Contribution for individuals under age 50	Maximum Contribution for individuals age 50 and older
under \$138,000	under \$218,000	\$0	\$6,500	\$7,500
\$139,500	\$219,000	\$1,000	\$5,850	\$6,750
\$141,000	\$220,000	\$2,000	\$5,200	\$6,000
\$142,500	\$221,000	\$3,000	\$4,550	\$5,250
\$144,000	\$222,000	\$4,000	\$3,900	\$4,500
\$145,500	\$223,000	\$5,000	\$3,250	\$3,750
\$147,000	\$224,000	\$6,000	\$2,600	\$3,000
\$148,500	\$225,000	\$7,000	\$1,950	\$2,250
\$150,000	\$226,000	\$8,000	\$1,300	\$1,500
\$151,500	\$227,000	\$9,000	\$650	\$750
\$153,000 & over	\$228,000 & over	\$10,000 & over	\$0	\$0



WHEN TO INVEST?

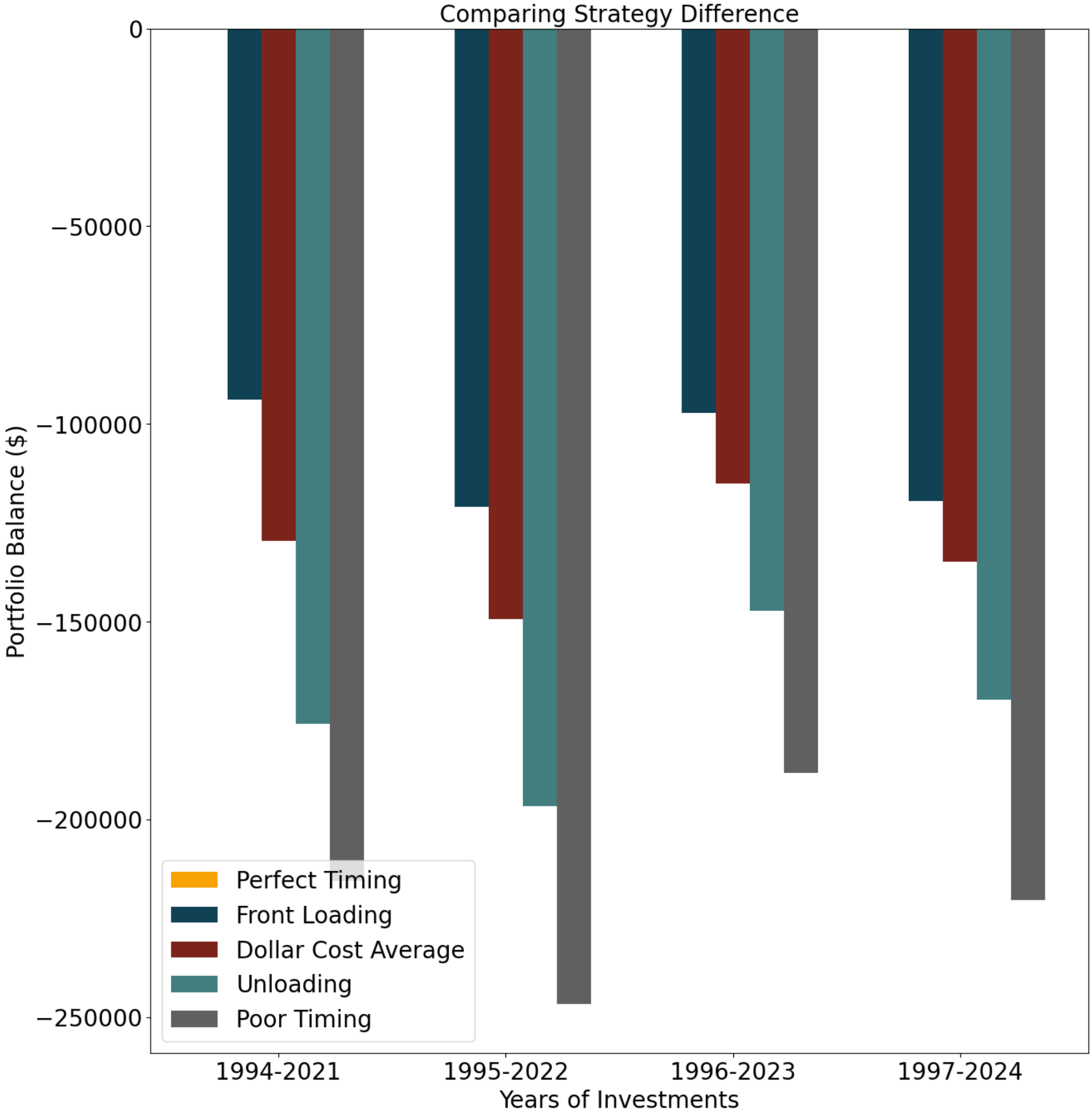
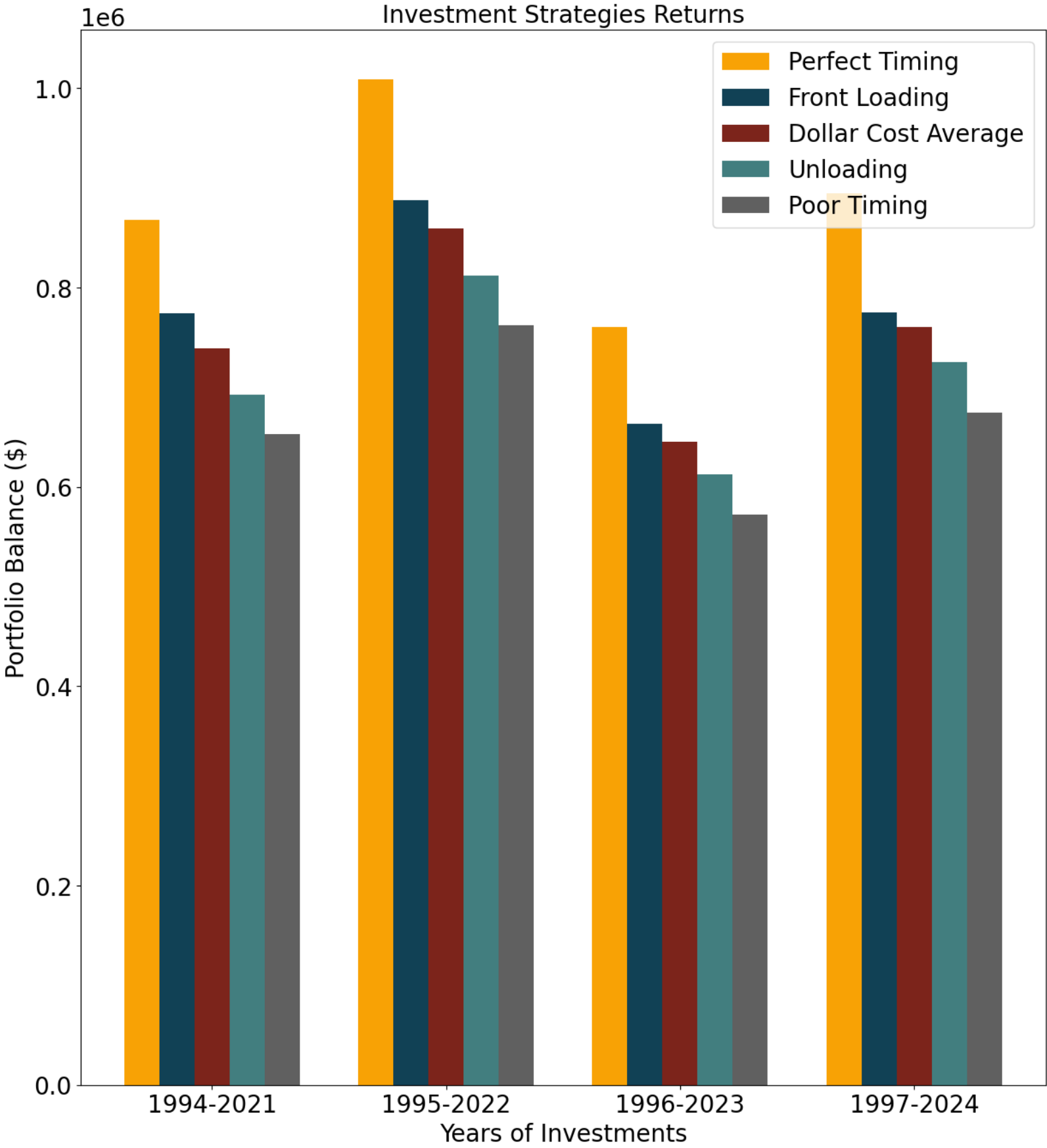
- **Front-loading investments means making a large initial investment upfront.** Front-loading can potentially yield higher returns due to earlier market exposure. This strategy is often used in retirement accounts, such as 401(k)s, IRAs, HSAs, where investors contribute the maximum amount early in the year.
- **Front-loading Drawbacks:**
 - Risk of missing out on employer matching contributions if contributions are completed early.
 - Less flexibility in adjusting contributions based on market conditions.
- **Dollar Cost Averaging on the other hand, involves spreading out investments over time, contributing smaller amounts regularly rather than a lump sum.** Reduces the risk of investing a large amount just before a market downturn. Allows for adjustments based on market performance and personal financial situations.



- **Dollar Cost Averaging Drawbacks:**
 - Potentially less growth compared to front-loading if the market rises significantly after the initial investment period.
 - May result in higher overall fees if investing in funds with front-end loads.

TIMING THE MARKET VERSUS TIME IN THE MARKET

- Investing \$7000 for 28 years in the S&P 500. Shifted the start and end year to get more samples.
- Front Loading is best option followed by DCA.
- Poor timing results in large losses
- 73% of the time the market are up. 27% of the time the market are down.



UNDERSTANDING INVESTMENT TERMS

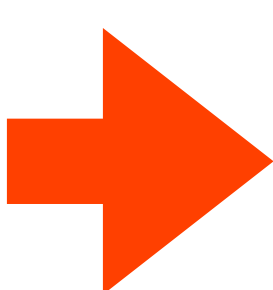
VGT ←

Vanguard Information Technology ETF

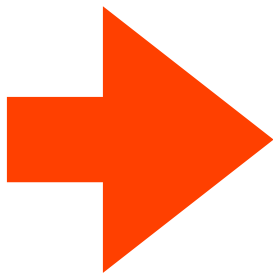
Also available as an [Admiral™ Shares](#) mutual fund. ←

Buy

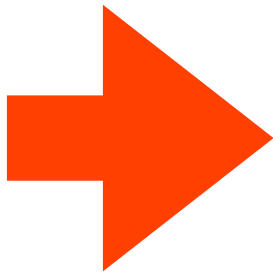
Compare



Management style
Index



Category
Technology



Expense ratio
0.09%
as of 02/03/2025

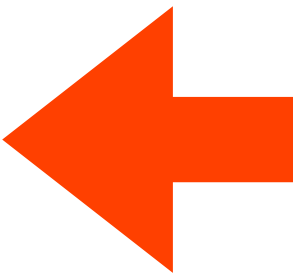
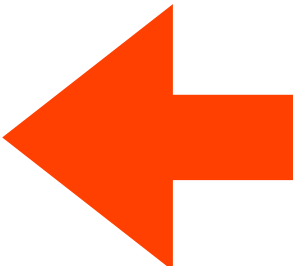
Market price (MP)
\$746.93
as of 11/26/2025

Asset class
Domestic
Stock -
Sector-
Specific

Risk / reward scale
Less  5 More

30 day SEC yield
0.36% B
as of 10/31/2025

NAV price
\$746.88
as of 11/26/2025



EXPENSES

How are expenses on ETF calculated

Management Fees	0.09%
12b-1 Distribution Fee	None
Other Expenses	0.01%
Total Annual Fund Operating Expenses	0.10%

Example

The following example is intended to help you compare the cost of investing in the Fund's ETF Shares with the cost of investing in other funds. It illustrates the hypothetical expenses that you would incur over various periods if you were to invest \$10,000 in the Fund's shares. This example assumes that the shares provide a return of 5% each year and that total annual fund operating expenses remain as stated in the preceding table. You would incur these hypothetical expenses whether or not you were to sell your shares at the end of the given period. Although your actual costs may be higher or lower, based on these assumptions your costs would be:

1 Year	3 Years	5 Years	10 Years
\$10	\$32	\$56	\$128

	Balance_No	Expense	Balance_Add	Expense	Diff
0		10500.00		10489.50	10.50
1		21682.50		21649.80	32.70
2		33583.99		33516.11	67.88
3		46242.81		46125.36	117.45
35		1224802.98		1198113.79	26689.19
36		1303989.09		1274689.48	29299.61
37		1387403.70		1355282.47	32121.23
38		1475262.27		1440093.44	35168.83

EXPENSES

- The gain assuming a fixed maximum contribution of \$25,000 over 30 years varies according to the investment expense in a non-taxable account.
- Column “FV” is the Future Value assuming no expense, “FV (1.0)” is the future value after 1.0% compound expense and “1.0” is the compound expense paid. That is, the expense paid \$328,893.03 will result in \$810,655.19 less gain over 30 years.
- Due to the compounding nature of interest, bigger percentage fees result in lesser gains

No Expense	Balance 0.1% Expense	Diff	No Expense	Balance 1.0% Expense	Diff	No Expense	Balance 2.0% Expense	Diff
26250.00	26223.75	26.25	26250.00	25987.50	262.50	26250.00	25725.00	525.00
54206.25	54124.51	81.74	54206.25	53391.32	814.93	54206.25	52581.90	1624.35
83959.97	83790.27	169.70	83959.97	82273.25	1686.72	83959.97	80609.31	3350.66
115607.02	115313.41	293.61	115607.02	112697.61	2909.41	115607.02	109847.06	5759.96
1678992.11	1652329.84	26662.27	1678992.11	1433420.70	245571.41	1678992.11	1229268.88	449723.23
1802180.22	1772410.65	29769.57	1802180.22	1528886.94	273293.28	1802180.22	1303371.42	498808.80
1932116.31	1898957.41	33158.90	1932116.31	1628706.78	303409.53	1932116.31	1380199.73	551916.58
2069146.62	2032295.44	36851.18	2069146.62	1733060.94	336085.68	2069146.62	1459841.52	609305.10

EXPENSES

- Table of returns after fees have been applied, starting from no fee to 3 % fee.
- The moral of the history is: “Watch out for those small fees”
- My mom use to say: “Learning does not cost you anything, not learning costs you a lot”

Year	0.0	0.035	0.04	0.08	0.1	0.3	0.8	1.0	1.5	2.0	3.0
1.0	25,300.00	25,291.15	25,289.88	25,279.76	25,274.70	25,224.10	25,097.60	25,047.00	24,920.50	24,794.00	24,541.00
2.0	53,130.00	53,101.67	53,097.62	53,065.25	53,049.07	52,887.37	52,484.10	52,323.18	51,921.86	51,521.93	50,726.25
3.0	83,743.00	83,682.54	83,673.90	83,604.84	83,570.32	83,225.68	82,368.25	82,026.95	81,177.84	80,334.64	78,665.91
4.0	117,417.30	117,309.72	117,294.35	117,171.51	117,110.13	116,497.70	114,977.84	114,374.34	112,876.69	111,394.74	108,477.52
27.0	3,063,828.53	3,043,281.92	3,040,358.74	3,017,081.16	3,005,513.95	2,892,419.68	2,629,197.48	2,531,240.97	2,303,213.65	2,097,358.44	1,743,591.67
28.0	3,395,511.38	3,371,729.59	3,368,346.73	3,341,414.00	3,328,033.98	3,197,340.76	2,894,077.89	2,781,568.42	2,520,452.49	2,285,746.40	1,884,953.31
29.0	3,760,362.52	3,732,895.58	3,728,989.21	3,697,894.72	3,682,451.24	3,531,747.71	3,183,115.40	3,054,175.01	2,755,830.77	2,488,828.62	2,035,786.18
30.0	4,161,698.77	4,130,039.12	4,125,537.26	4,089,709.80	4,071,920.37	3,898,491.81	3,498,513.12	3,351,043.58	3,010,863.14	2,707,751.25	2,196,724.85

<https://www.wpcu.coop/helpful-resources/financial-calculators>

<https://www.calculator.net/financial-calculator.html>

TAX EFFICIENCY

- ROTH, HSA and 529b are more tax efficient than a Traditional, FSA or Brokerage Account
- Money Market (specifically Vanguard) are more tax efficient than a CD, Dividend ETF or a High Yields accounts on which you are getting taxes every year for every single dollar.
- ETF with low dividend are more efficient than High Dividend ETF, Mutual Fund or Individual stocks
- If you are planning to have money in reserve, a VSUXX most prudent place to hold the money at the moment (2025).
- If you want to save for children's education a 529b with a SP500 ETF is the best location (move it to VSUXX/VTMFX 5 years before needing the money)



Most Tax Efficient

- Low-yielding cash, money market
- Tax-managed stock funds
- Total-market stock index funds
- Mid-cap and small-cap index funds
- Active stock funds
- Balanced funds
- Taxable bond funds
- REIT funds
- High-yield bond funds

Less Tax Efficient

THE POWER OF TAXES

- Many studies have been done to determine how federal taxes reduce mutual fund returns to fund shareholders. One of the longest studies, commissioned by Charles Schwab, was for the 30-year period from 1963 to 1992. The study found that a high-bracket taxpayer who invested \$1.00 in U.S. stocks at the beginning of that period would have \$21.89 at the end of the period if invested in a tax-deferred account. Meanwhile, the same \$1.00 would have grown to only \$9.87 if invested in a taxable account—less than half as much. These figures clearly show that taxes, plus the impact of compounding, make a tremendous difference in after-tax performance
- For maximum tax efficiency in taxable accounts, you should do the following:
- Favor funds with low dividends.
- Favor funds with “qualified” dividends.
- Favor funds with low turnover.
- Favor tax-efficient index funds and tax-managed funds.
- **You can control the expenses and the taxes**

TYPES OF MARKET RISK

Market risk refers to the potential for losses in investments due to fluctuations in market variables.

- 1. Equity Risk:** The risk of loss due to changes in stock prices or their volatility. You investment losing value when you need it like are retirement age or close to buying a home.
- 2. Interest Rate Risk:** The risk associated with changes in interest rates. Defaulting, reducing the rate or extending the pay period. REITs, High Yield (Junk Bonds) are a common high interest rate risk investments.
- 3. Currency Risk (Foreign Exchange Risk).** The risk of loss due to fluctuations in exchange rates.
- 4. Commodity Risk:** The risk of loss due to changes in commodity (oil, gold, home prices, and agricultural products) prices.

5. Opportunity Risk: While not a traditional market risk, refers to the potential loss of benefits from not pursuing an alternative investment or strategy.

6. Shape Risk: The risk related to changes in the shape of the yield curve. Influences the pricing of interest rate derivatives.

7. Holding Period Risk: The risk associated with the time an investment is held. Longer holding periods can increase exposure to market volatility.

8. Basis Risk: The risk that the price of a hedging instrument does not move in line with the price of the asset being hedged. Affects the effectiveness of hedging strategies.

9. Margining Risk: The risk of uncertain future cash outflows due to margin calls. Impact: Affects leveraged positions in trading.

UNDERSTANDING THESE RISKS HELPS MAKE INFORMED DECISIONS AND MANAGE PORTFOLIOS EFFECTIVELY.

DIVIDEND / HIGH YIELDS ETFS VS EQUITY ETFS

- Reduced Capital Appreciation Potential

- Focus on Mature Companies: Dividend ETFs, especially those with high yields, tend to hold stocks of mature, stable companies (often in sectors like utilities, financials, and consumer staples) that generate significant cash flow. These companies distribute a portion of their profits as dividends rather than reinvesting all of it back into the business for high-speed expansion.
- Less Exposure to High-Growth Firms: Pure growth companies (often in technology or biotech) typically reinvest almost all of their earnings back into research, development, and expansion to achieve higher revenue and profit growth. They pay little to no dividend, which is why they are often excluded from or have a lower weighting in dividend ETFs.

- Sector Concentration Risk

- Bias Towards Certain Sectors: To achieve their dividend mandates, these ETFs often have an overweighting in sectors known for stable, consistent cash flow and high dividend payouts (like Utilities and Consumer Staples).
- Limitation on Growth Sectors: Conversely, they often have an underweighting or even exclusion of high-growth sectors (like Technology) that are the traditional drivers of major market rallies.
- The Growth "Cost": This concentration means your portfolio may miss out on significant growth driven by innovation and secular trends in the most dynamic sectors of the economy.

**COMPARE VGT, SCHD AND VIG IN
VANGUARD ETC COMPARE**

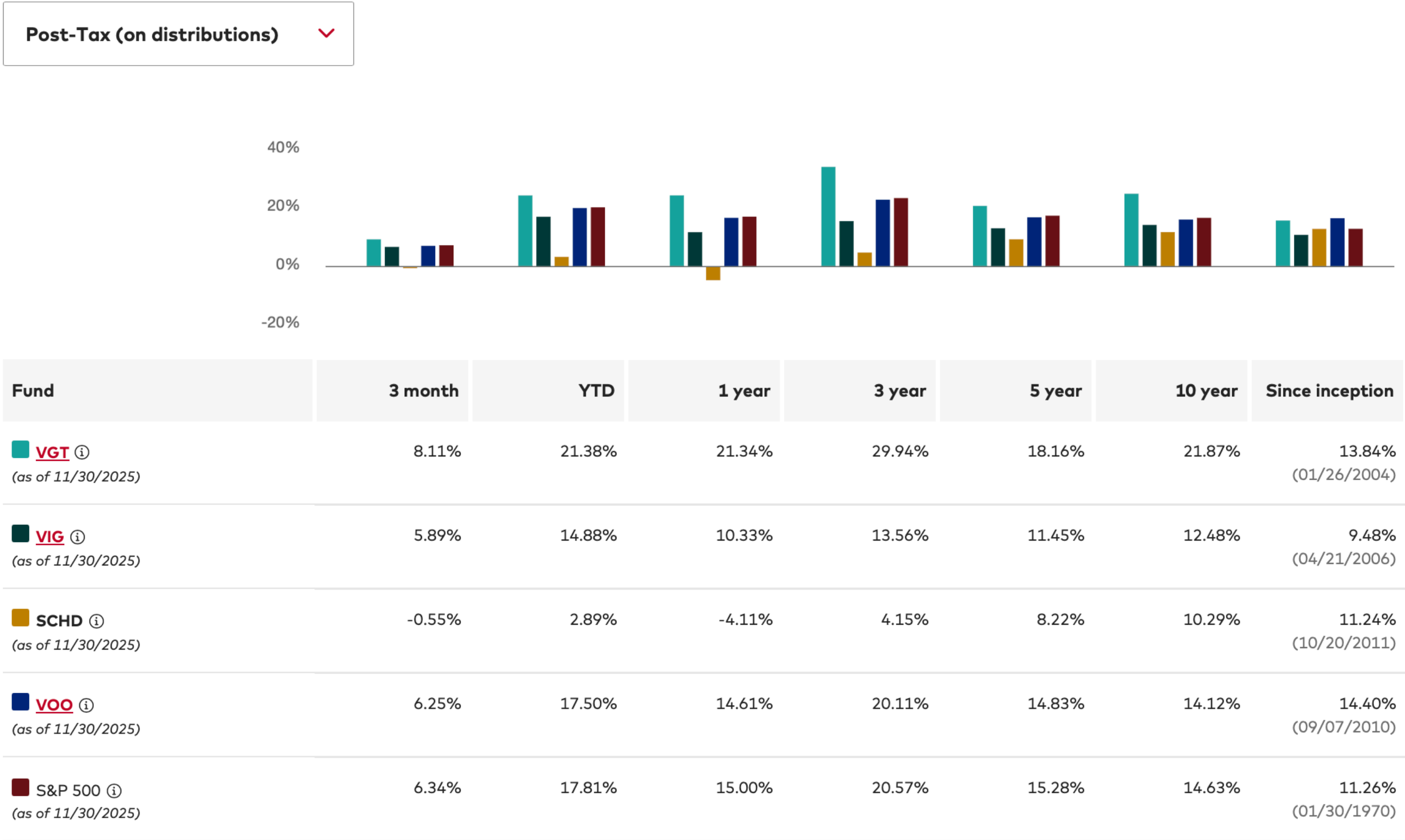
**COMPARE VGT, SCHD AND VIG IN ETF
INSIDER**

**COMPARE VGT, SCHD AND VIG IN ETF
RESEARCH CENTER**

**COMPARE VGT, SCHD AND VIG IN
PORTFOLIO VISUALIZER**

DIVIDEND / HIGH YIELDS ETFS VS EQUITY ETFS

- **Taxable Events:** Unlike growth stocks, where you don't pay capital gains tax until you sell the stock, the dividends you receive from a dividend ETF are generally taxed each year.
- Qualified dividend can be taxed at a capital gains tax rate of 0%, 15%, or 20%.
- Non-Qualified Dividends are taxed as ordinary income (a higher rate). If you don't hold the shares long enough, the IRS might deem them nonqualified, and you'll pay the higher, nonqualified tax rate.
- **The Cost:** This annual taxation means the power of compounding is slightly reduced, as a portion of your return is diverted to taxes each year instead of being fully reinvested (unless the ETF is held in a tax-advantaged account like a Roth IRA or 401k).



Pre-Tax Returns shown are net of fees and expenses and include the reinvestment of dividends and interest income. The performance shown does not include taxes owed on distributions. **Post-Tax Returns are net of fees and expenses and do adjust for taxes and sales charges.**

DIVIDEND AND HIGH YIELDS ETFs

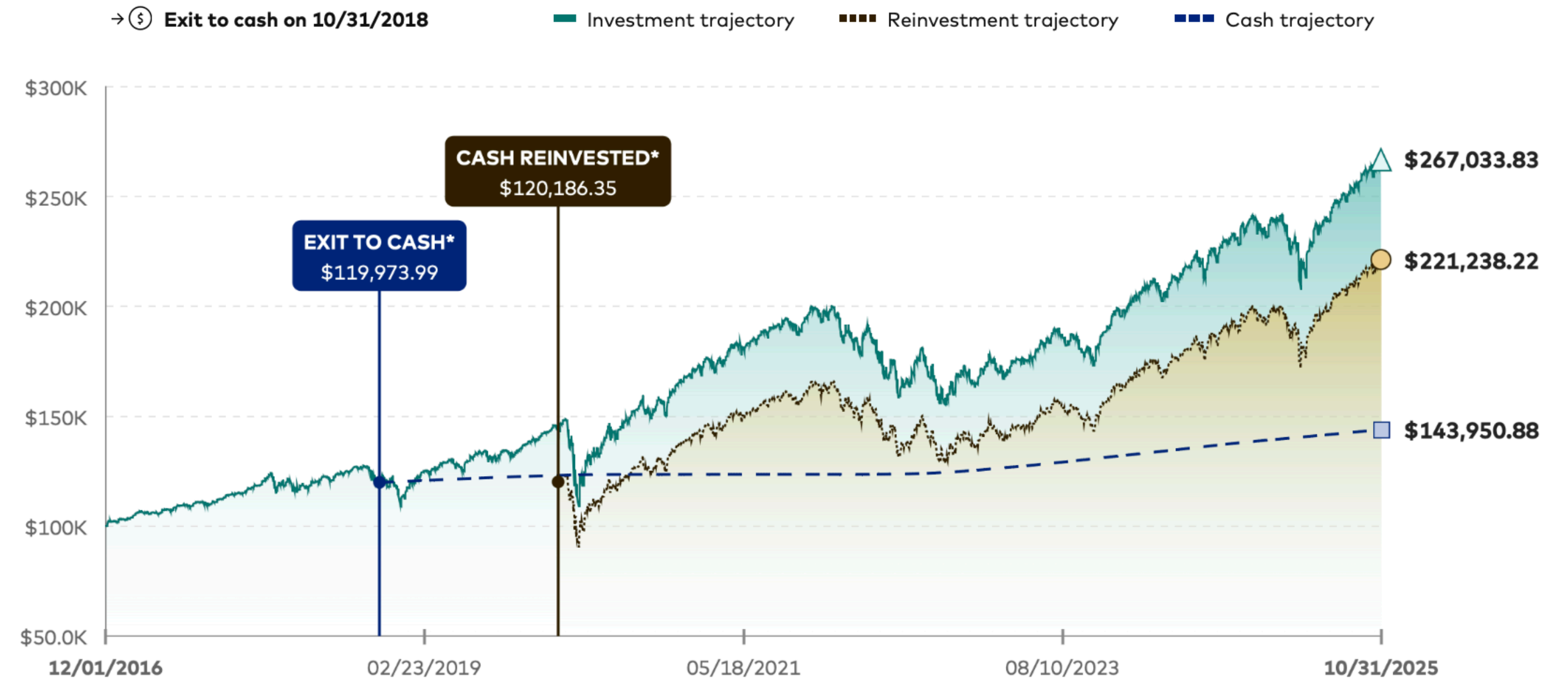
- If you are interested in growth. High dividend and high yield ETFs are not the way to go.
- The reason is the tax implication of dividend are almost always taxed at 15%rate while
- The second reason is that at retirement you can to control where you take money from for tax efficiency. If you have 1M and getting 3% which is \$30k you are screwed in taxes. While could have taken that money out the 401k that required the minimum distribution.
- It is a bad idea to put a Tax Inefficiency investment in Non-Tax Sheltered account

Tax cost ratio ⓘ					Hide ^
Fund	1 year	3 year	5 year	10 year	
VGT ⓘ (as of 11/30/2025)	0.11%	0.17%	0.20%	0.26%	
VIG ⓘ (as of 11/30/2025)	0.42%	0.45%	0.53%	0.54%	
SCHD ⓘ (as of 11/30/2025)	1.36%	1.24%	1.15%	1.00%	
VOO ⓘ (as of 11/30/2025)	0.31%	0.35%	0.35%	0.41%	

The Morningstar Tax Cost Ratio measures how much a fund's annualized return is reduced by the taxes investors pay on distributions. Mutual funds regularly distribute stock dividends, bond dividends and capital gains to their shareholders. Investors then must pay taxes on those distributions during the year they were received. Like an expense ratio, the tax cost ratio is a measure of how one factor can negatively impact performance. Also like an expense ratio, it is usually concentrated in the range of 0-5%. 0% indicates that the fund had no taxable distributions and 5% indicates that the fund was less tax efficient. **For example, if a fund had a 2% tax cost ratio for the three-year time period, it means that on average each year, investors in that fund lost 2% of their assets to taxes. If the fund had a three-year annualized pre-tax return of 10%, an investor in the fund took home about 8% on an after-tax basis. (Because the returns are compounded, the after-tax return is actually 7.8%.**

THE POWER OF EMOTIONS

- Don't fall for the Cramers yelling to sell. They are pushing you to lock in the loss so they can lock in the wins ("market correction" or "pricing in losses")
- "In every market, bulls win, bears win and pigs get slaughtered." - Bloomberg Guest Speaker
- Emotions like fear and greed significantly influence stock market behavior, often leading to poor investment decisions. **Maintaining a long-term mindset helps investors avoid emotional reactions to market fluctuations, allowing them to focus on their overall financial goals instead.**

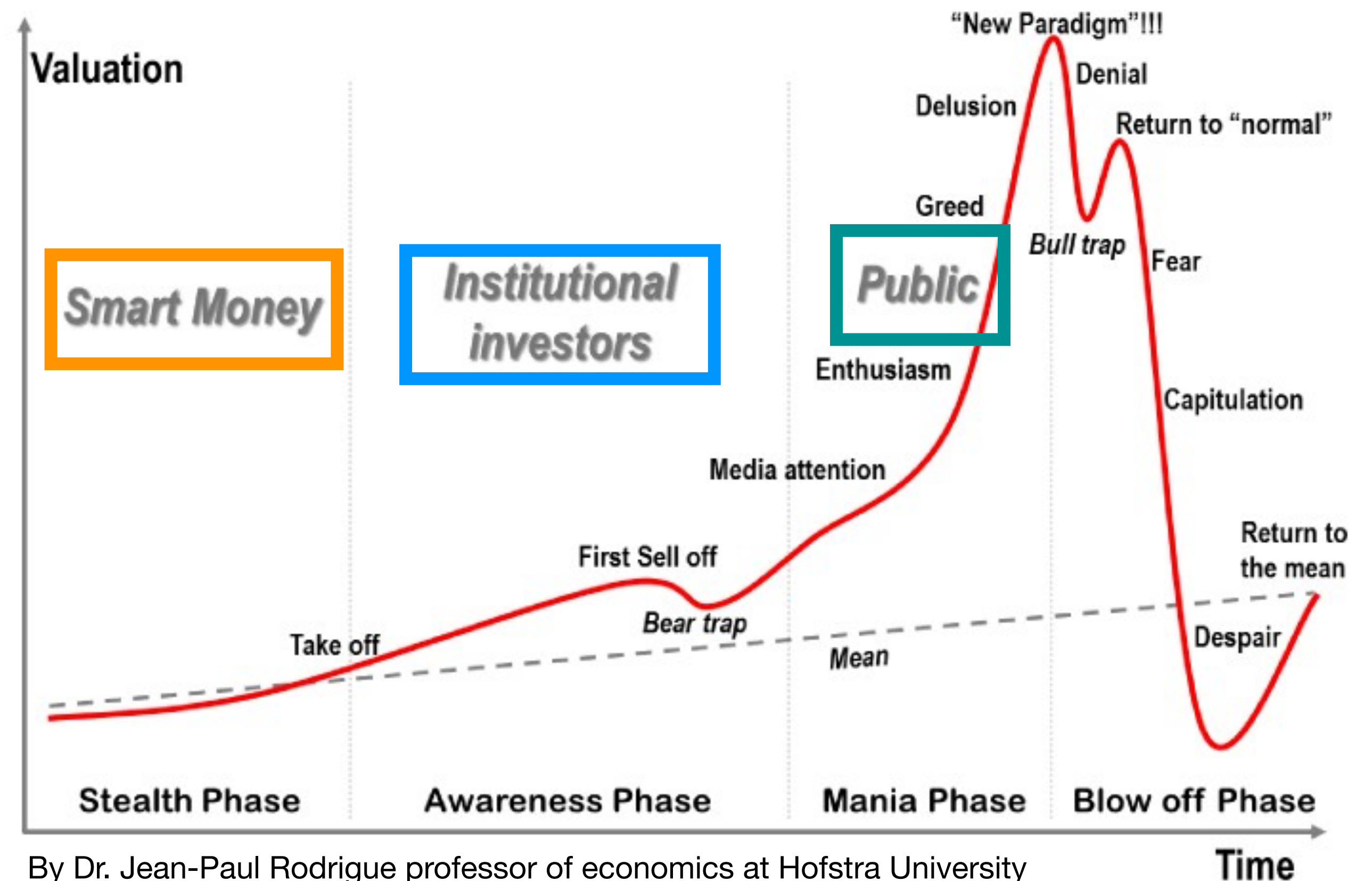


- ▲ **Staying invested** resulted in a portfolio ending balance of **\$267,033.83**, a **\$123,082.95** gain in portfolio value.
- **Selling to cash** resulted in a portfolio ending balance of **\$143,950.88**, a **\$123,082.95** loss in value compared to remaining invested.
- **Reinvesting cash** resulted in a portfolio ending balance of **\$221,238.22**, a **\$45,795.61** loss in value compared to remaining invested.

THE POWER OF EMOTIONS

- Psychological stages of the stock market boom and bust cycle
- A market bubble is a run-up in stock prices without a corresponding increase in the value of the businesses they represent. In a bubble, speculation and euphoria take over.
- “Everyone is an expert during a bull market”
- These cycle repeat all the time so...
 - Think long term, not short-term
 - Improve your portfolio (Diversification)
 - Buy on sale (Low P/E ratio)

Psychological stages of the stock market boom and bust cycle



HOW TO COMPARE INVESTMENT PRODUCTS

- Dont only look at the returns
- Compare expense ratios (lower is better)
- Look at the P/E overtime (lower is better)
- Look at the distribution over time for tax purposes
- 10yr avg return is better than all time avg return statistics due to difference in fund start date
- Compare turn-over ratio (lower is better)
- Compare tax cost ratio (lower is better)

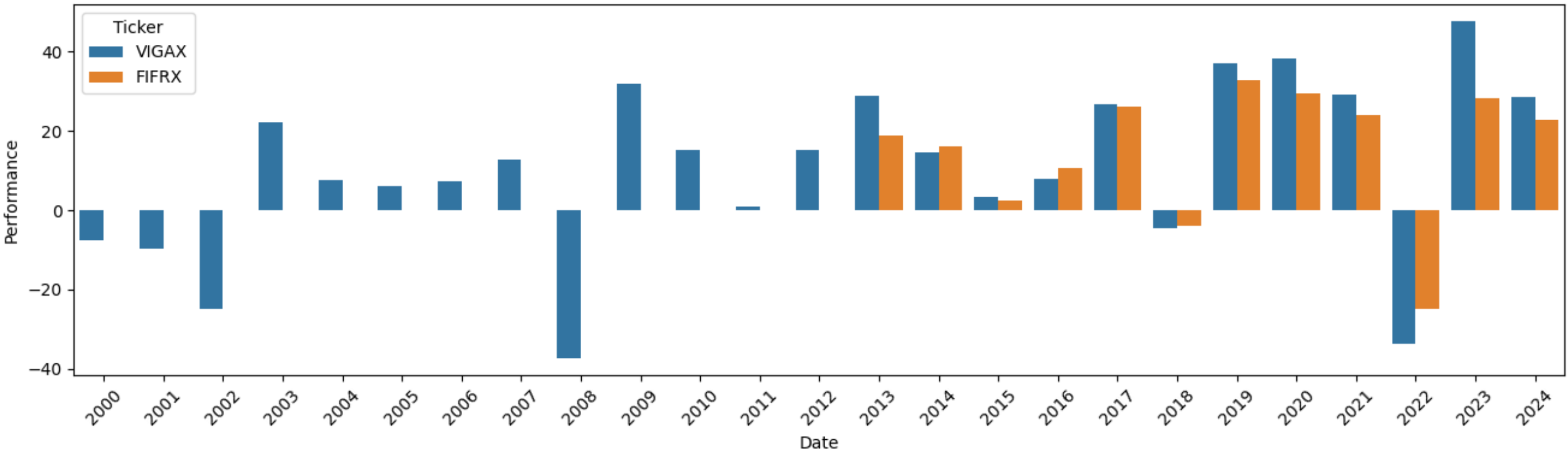
Comparing Averages

	VIGAX	FIFRX
YTD	28.71	22.85
1yr	38.25	25.58
3yr	14.3	8.8
5yr	22.1	15.99
10yr	18.13	14.82
All	10.6	15.27
+ Yrs	19	10
- Yrs	6	2

```
df1 = fun(years,yearly_deposit,13.55,present_value,.47,inflation,age)
df0 = fun(years,yearly_deposit,15.54,present_value,.05,inflation,age)
df0 = df0.merge(df1, on='Age', suffixes=('_VIGAX', '_FIFRX'))
df0.iloc[[0,1,2,3,-4,-3,-2,-1]]
```

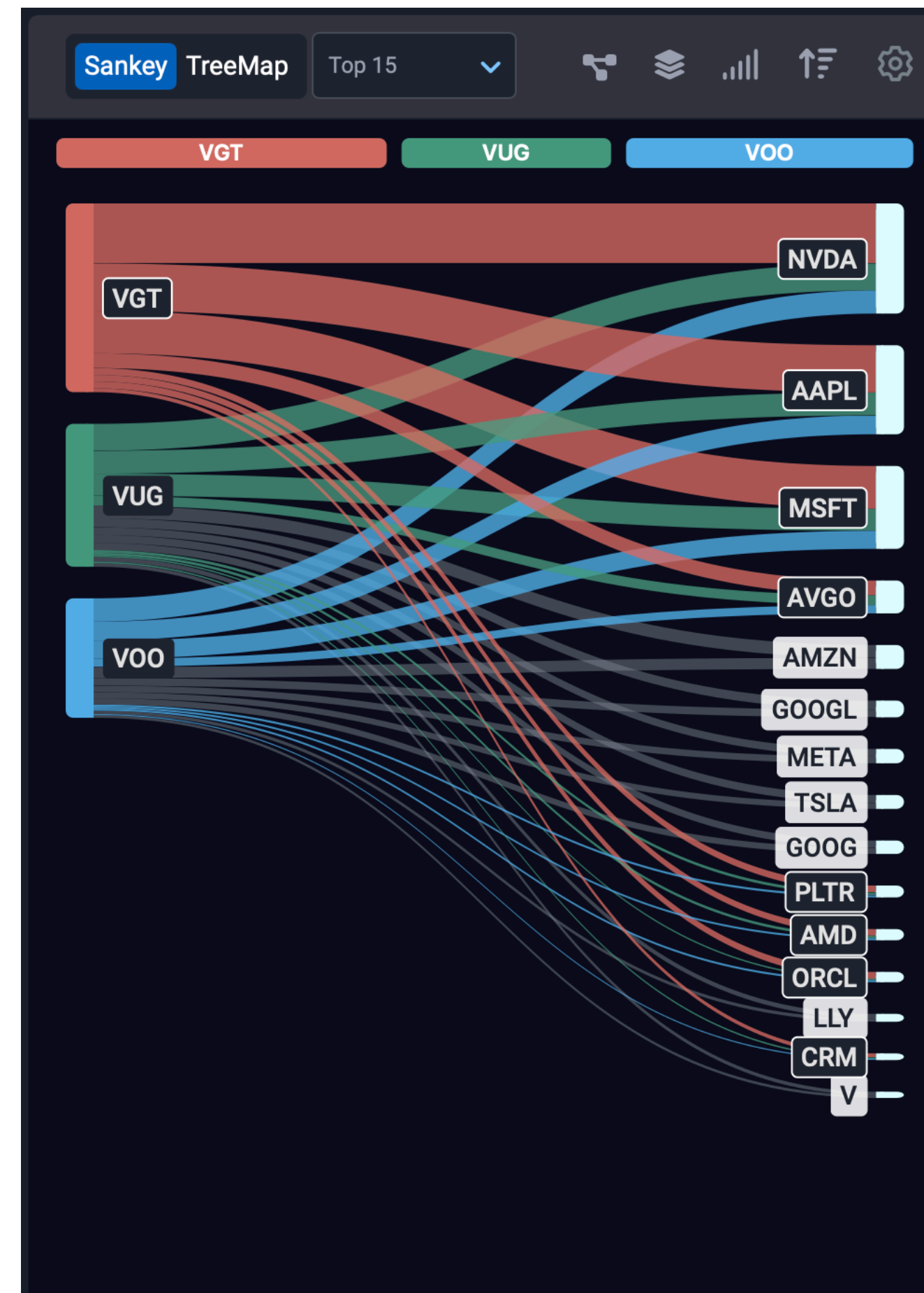
Including Expenses and Inflation

	Age	Balance (Inf. Abj.)_VIGAX	Balance (Inf. Abj.)_FIFRX
0	37.0	343,559.63	336,223.54
1	38.0	468,621.27	450,642.61
2	39.0	605,395.99	572,931.52
3	40.0	755,227.33	703,821.82
27	64.0	16,427,430.88	10,042,497.10
28	65.0	18,286,289.12	10,957,340.59
29	66.0	20,349,290.74	11,950,434.11
30	67.0	22,639,000.52	13,028,587.11

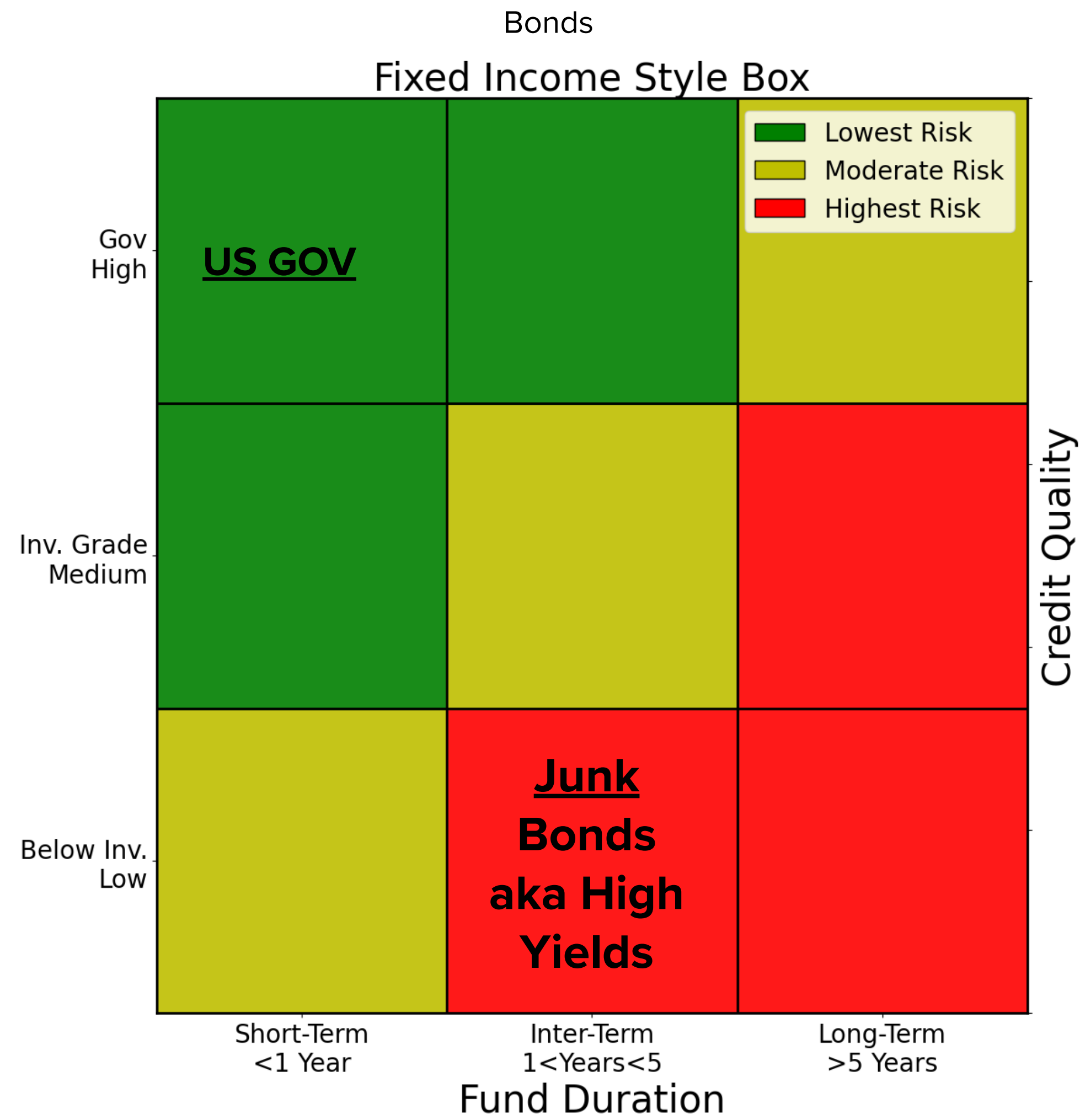
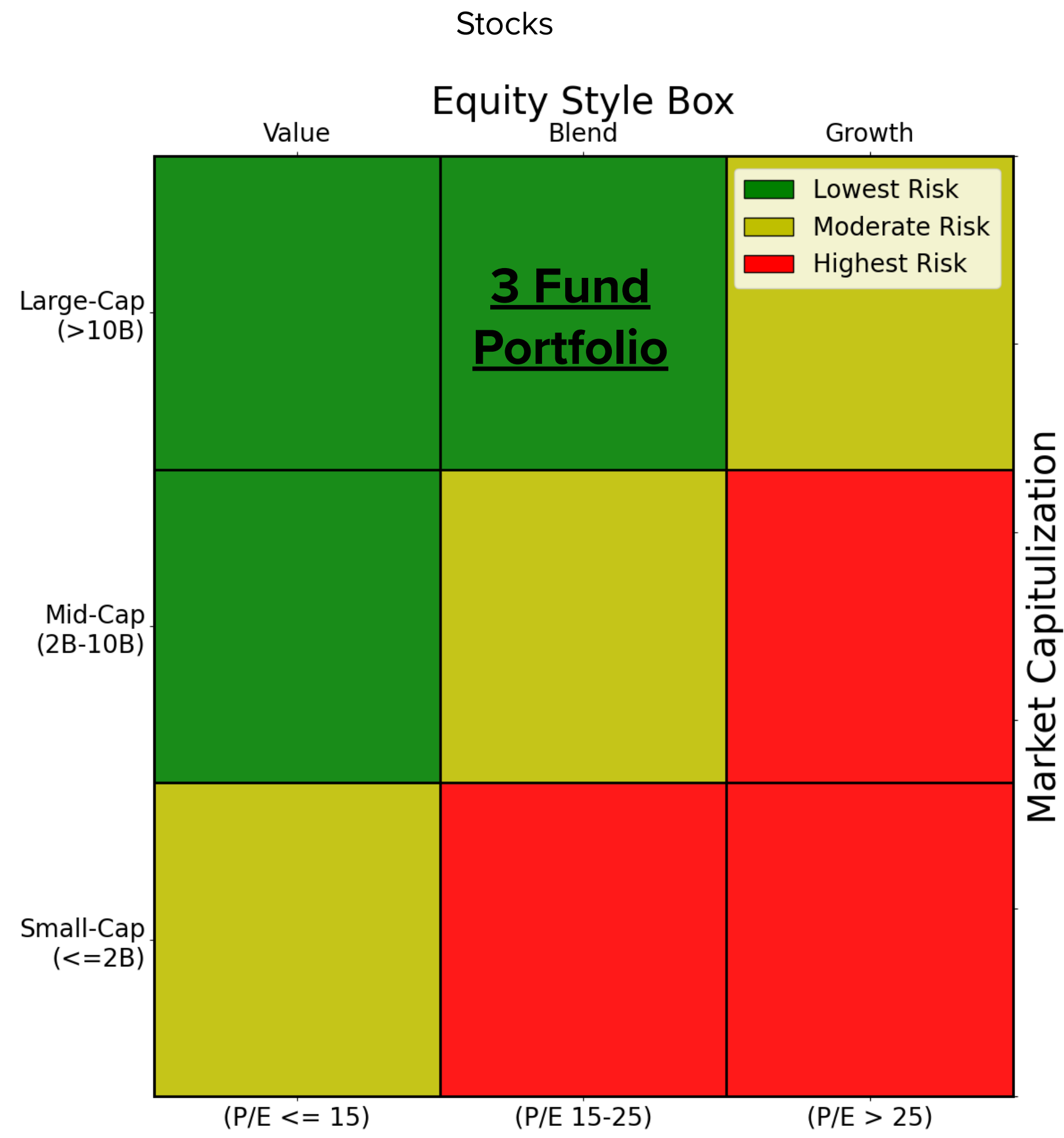


PORTFOLIO OPTIMIZATION AND DIVERSIFICATION

- Find matching funds for tax loss harvesting
- Find mutual excluding funds for diversification
- Customize portfolio for market concentration
- 51% of the total investment between VGT and VUG is in the same companies
- Custom ETF Overlap Calculator



HOW TO COMPARE INVESTMENT PRODUCTS



COMMON MISTAKES

1. Listening to TikTok Financial Influencers
2. Not having the immediate access to accounts (HSA, IRA, 401k)
3. Security is super important (weak password, using android or no multi-factor authentication)
4. Selling in a down turn (Do not sell when markets are readjusting, that is what they want to “price in losses”)
5. You shall not buy the peak on a pump and dump meme stock
6. Not looking at the fundamentals (P/E) and expense but only the stock value
7. Not taking into account the taxes in a taxable account
8. Chasing high yield / junk / high dividend stocks (resulting in higher taxes and less returns with added risk)
9. Not diversifying. High risk investments the first 10 years of investing
10. Timing the market, delayed investment, unnecessary dollar cost averaging ROTH IRA and HSA.
11. Not investing HSA or ROTH IRA (Leaving in settlement account)
12. Not having a cash management account
13. Thinking that past performance is indicative of future performance (This is not basketball)
14. Not taking advantage of all the free information out there including this one :P
15. Not knowing the minimum or maximum contribution limits for the current year
16. The following the investment order as prescribed by Financial Order of Operation (FOO)
17. No awareness of Fidelity, Vanguard, and others investment products (VOOG vs RGAGX vs FSPGX vs Russel 1000 Growth)

THINGS TO WATCH OUT FOR

- Market concentration
- High Expenses
 - Active Managed ETF
- Tax Inefficiencies
 - Dividend and High Yields
- Complex portfolios
 - Dozens of ETFs
- Exceeding Contribution Limits
 - Income Increase, bonus, work change
- Not leveraging taxable and tax sheltered accounts

GOOD RESOURCES

- Useful books to read

1. The Simple Path to Wealth
2. Exchange-Traded Funds For Dummies, 3rd Edition
3. Bogleheads' Guide to Investing
4. The Little book of Common Sense Investing
5. Youtube channels: Rob Berger, Ben Felix, Erin Talks Money, The Plain Bagel, Coffeezilla,